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Applied Machine Learning

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Early Estimation of District Paddy Yields in Sri Lanka

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A Machine Learning Approach Using Official Crop Statistics, Satellite Rainfall and Weather Data

Code and data github.com/Uthpalagit/paddy-yield-prediction-269219J

Executive Summary

District paddy yields in Sri Lanka are confirmed only when the Department of Census and Statistics completes its crop-cutting survey, several months after each harvest. This project developed a model that estimates the yield of each district as soon as a season ends. A dataset of 2,276 district-seasons (1981-2026) was assembled from official crop statistics, district-averaged satellite rainfall and weather records, and subjected to extensive cleaning and validation.

Twenty modelling approaches were compared on a strictly time-based design. The selected model, an Explainable Boosting Machine, estimated yields for the unseen 2017-2026 seasons with an accuracy of 87.6% (average error 446 kg/ha, or 12.4%), against 532 kg/ha for the best naive benchmark. 86% of its district estimates fell within 20% of the official figure, and the national production it implies was within 5.8% on average outside the fertilizer crisis. Every estimate can be decomposed into the contribution of each input.

No model anticipated the collapse of 2021/22 following the fertilizer import ban, and none of the additional sources tested, including climate indices, the subsidy record, satellite crop health and soil moisture, revealed it in advance. Once one shortage season has been observed, however, a policy input enables the model to recognise the next, and the deployed model therefore includes a fertilizer-shortage scenario. The model is delivered through an interactive web application and a containerised prediction service with automated testing and drift monitoring.

1. Background and Objective

Rice is the staple food of Sri Lanka and is cultivated in two seasons each year. *Maha* is the main season from approximately October to March, and *Yala* is the minor season from approximately April to September. The official average yield of each district is published only after the Department of Census and Statistics (DCS) completes its crop-cutting survey. Until then, the Department of Agriculture, the Paddy Marketing Board and the relevant ministries plan procurement, storage and imports without reliable yield information. The cost of this delay became evident in the 2021/22 Maha season, when the national yield fell by approximately one third following the ban on chemical fertilizer imports, and rice had to be imported at short notice.

The objective of this project was to estimate the average paddy yield (kg/ha) of each district as soon as a season ends, using only information available at that time, namely the extent cultivated and its irrigation type, the district's own yield history, and the weather during the season. A secondary objective was to explain each estimate, so that planners can judge when to rely on it.

2. Data Sources

No suitable ready-made dataset was available, so the modelling dataset was constructed from public sources. Many more sources were collected than were finally used. The aim was to find the best mix of features, and every candidate was tested before it was accepted (Section 4.2). The sources are listed below.

- Department of Census and Statistics, Paddy Statistics [1]. Eight tables of average yield, sown extent, harvested extent and production for every district and season since 1978/79, with extents broken down by major irrigation, minor irrigation and rainfed cultivation.
- CHIRPS rainfall [2]. Daily satellite and rain-gauge rainfall at a resolution of approximately 5 km, averaged over the boundary of each district through the NASA/USAID ClimateSERV service, for 1981-2026.

- NASA POWER [3]. Monthly temperature, humidity, solar radiation and wind for each district.
- OpenStreetMap. District coordinates and boundaries.
- FAOSTAT [4] and NASA MODIS. National fertilizer imports and satellite vegetation greenness (NDVI). Both were evaluated as additional inputs (Section 4.2) but were not retained.
- NOAA climate indices and crop health [12, 13]. The El Niño (ONI) and Indian Ocean Dipole (DMI) indices, and weekly vegetation health indices averaged over the rice-growing area of each district (1982-2026). These were also evaluated and not retained.
- Government fertilizer policy [14, 15]. The subsidy regime in force in each season since 1981, from the published policy record, including the import ban of May 2021 and the shortage of 2022.
- ERA5-Land soil moisture [16]. Daily root-zone soil moisture for each district from 1980, used with NASA POWER to derive evapotranspiration and stand-ins for reservoir storage and heat stress. These were also evaluated and not retained.
- Provinces and agro-ecological zones. The province and the Department of Agriculture climate zone of each district, evaluated as qualitative features and not retained (Section 4.3).

The unit of analysis is the district-season. The 27 units comprise the 25 administrative districts and two irrigation project areas (Mahaweli System H and Udawalawe) that the DCS reports separately.

3. Data Preparation and Quality Assurance

The official tables required substantial cleaning. Every issue was identified through an explicit check, corrected, and the correction verified before proceeding. The principal findings are listed below.

- Inconsistent structure. The header row and column differed between files, and irrigation categories were spelt inconsistently. A cell count confirmed that the reshaping retained all 34,646 data cells.
- Inconsistent district names. 34 spellings were found for 27 units, together with a national total and a separate highland category, both of which were removed.
- Mislabelled seasons. The 2020/21 Maha column was labelled "2000/2021" in three tables. Read as recorded, it would have overwritten the 2000/01 season. A year recorded as "198" was corrected to 1980.
- Impossible values. 21 yields were recorded as exactly zero, almost all in Yala seasons of 1988-1994 in the Northern and Eastern provinces, although the same records show up to 37,000 ha harvested. These reflect seasons in which the survey could not be conducted during the conflict and were treated as missing. One harvested extent exceeded the sown extent by 87% and two irrigation breakdowns did not sum to the total. These values were also set to missing.
- Inconsistent tables. The production table did not reconcile with harvested extent multiplied by yield and was not used, as it is in any case derived from the target.
- Weather resolution. NASA POWER provided only 19 distinct weather series for the 27 units, because neighbouring districts fall in the same grid cell. This finding led to the adoption of district-averaged CHIRPS rainfall, which provides a distinct series for every unit.
- Outliers. Extreme but genuine seasons, such as the 2021/22 collapse and record yields in irrigated areas, were retained. Six yields measured on less than 100 ha were excluded from model training.
- Placeholders, duplicates and types. 6,130 cells recorded as a dash and the NASA code -999 were converted to missing values. No duplicate district-seasons were found, and every column was converted to its proper type.

Weather was aggregated over each season's growing window according to the Department of Agriculture crop calendar. Because CHIRPS is daily, two further indicators could be derived, namely the longest dry spell during the growing months and the number of days with at least 50 mm of rain. History features (the previous same-season yield, the three-season average, the previous season and an ARIMA trend forecast) were computed only from earlier seasons to prevent information leakage.

The remaining 3,372 gaps in the inputs were filled with the median of the training seasons inside the modelling pipeline, never on the whole table, and a missing yield was never filled. For the models that require it, numeric inputs were standardised and district and season were one-hot encoded, while CatBoost and the EBM used the cleaned columns natively. A log transform of five skewed features (358 against 355 kg/ha) and a robust scaler (355.2 against 355.0) gave no gain, and correlation and VIF checks led to one of three temperature measures being kept.

Exploratory analysis showed that the rainfed share is the strongest single correlate of yield ($r = -0.61$, against -0.26 for heavy-rain days), that Maha exceeds Yala in 65% of district-years (by 136 kg/ha on average, $p < 0.001$), and that the yield distribution is close to symmetric (skew 0.4), so the target was modelled without transformation.

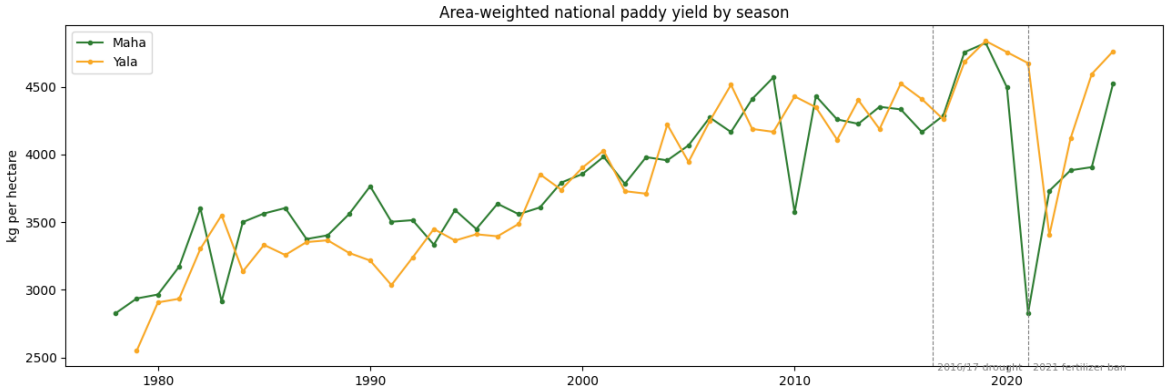


Figure 1. Area-weighted national yield by season, showing the long-term increase and the 2021/22 collapse to 2,828 kg/ha.

4. Methodology

4.1 Validation design

Because the data are ordered in time, a random split would allow the model to learn from the future. The data were therefore split by time. Models were trained on 1981-2016 and evaluated once on the 446 district-seasons from Yala 2017 to Maha 2025/26. All modelling decisions were made on seven yearly validation folds within the training period (training on all seasons before a given year and validating on that year, for 2010 to 2016). Preprocessing (median imputation, standard scaling and one-hot encoding of district and season) was performed inside pipelines fitted on training data only. CatBoost and the EBM, which handle categories and missing values themselves, received the cleaned columns directly, and training them through the same pipeline changed their validation error by less than 1%. The most recent Polonnaruwa season was held out entirely for a final check, and nine leakage checks passed in code.

4.2 Feature selection

Feature selection determined which inputs the model should use. Each candidate addition was tested individually with three different algorithms, held at fixed settings. Improvements to data quality were adopted where they did not increase the average validation error, while new external data sources were required to reduce it by more than 1%, consistent with the threshold used for model selection. On this basis, CHIRPS rainfall, the dry-spell and heavy-rain indicators, and a single temperature variable were retained. Nitrogen fertilizer imports, conditions in neighbouring districts, satellite greenness, the El Niño and Indian Ocean Dipole indices (+0.3% and -0.6%), the fertilizer subsidy level (-0.2%), crop health over the rice fields (-0.8%) and an alternative target formulation were not retained. Fertilizer imports were of particular interest, but in the training years import volumes fluctuated widely without corresponding changes in yield, so the 2021 reduction did not appear unusual to the model. Satellite crop health was equally uninformative about the crisis. In Maha 2021/22 the peak greenness of the rice fields (NDVI 0.400) was within the normal range of the decade, although yields fell to 61% of their three-season average.

Further candidates were tested after the main analysis was complete. These were soil moisture and evapotranspiration (+0.2%), stand-ins for reservoir storage and planting conditions (+0.6%), heat in the hottest month (0.0%) and weather at each growth stage, namely heat at flowering, night temperature and sunshine during ripening, and rain at harvest (-1.5%), and four qualitative features, namely the fertilizer policy regime, province, climate zone and El Niño phase (0.0%). The harvested share was also scored, only to illustrate leakage. It appeared 2.7% better because it is published together with the yield.

4.3 Further feature checks

To test whether some other combination of all 43 candidates would perform better, three standard feature selection methods were applied, namely Lasso [17], permutation importance [18] and backward elimination [19]. To stop the methods grading their own choice, they selected features on the 2010-2013 validation years and were judged on 2014-2016. These late checks, however, reused validation years that had already informed every earlier decision, and with enough attempts some options look better on the same years by chance, which is known as overfitting to the validation set. They were therefore reported but not permitted to change the model. On the test seasons, neither the growth-stage weather (447 kg/ha) nor the consensus of the three methods (464 kg/ha) improved on the final feature set (446 kg/ha), which supports that decision. A blend with ridge regression, weighting recent seasons and tuning the EBM each changed the error by about 1% or less, and principal component analysis of the weather features made it worse (by 0.2 to 1.8%). The qualitative features were the one addition that improved the EBM on both validation (341 against 347 kg/ha) and test (434 against 446 kg/ha), but the gain was small and unstable, so it is recommended for confirmation on the 2026/27 seasons rather than adopted. Of the eleven datasets loaded, the final model uses five, namely the DCS statistics, CHIRPS rainfall, NASA POWER weather, the fertilizer policy record and OpenStreetMap to locate each district. The other six were dropped completely.

4.4 Model selection and parameter engineering

Feature selection fixed the inputs before any algorithm was chosen, and every model was then compared on the same inputs and the same validation folds. Keeping the two steps apart keeps the number of combinations small and limits overfitting to the validation years. Twenty candidate approaches were compared. They comprised naive and ARIMA baselines, ridge regression and a decision tree, bagging and random forest, XGBoost, LightGBM and CatBoost, a stacking ensemble, two hybrids (ARIMA with a random forest on its residuals, and a ridge and CatBoost blend), and seven additional methods, selected for their suitability to the data. The seven were the Explainable Boosting Machine (EBM) [7], an additive model that learns one function per feature, a linear mixed-effects model with a random intercept for each district, NGBoost [8], which predicts a probability distribution, Gaussian process regression, extremely randomised trees, histogram gradient boosting and Huber regression. Hyperparameters were set with reference to the data and checked by early stopping and grid search. Two methods from published Sri Lankan studies [5, 6] were rebuilt separately as a benchmark (Section 6). They were not candidates for selection.

The selection rule, applied to validation results only, was the lowest mean absolute error (MAE). Models within 1% of the best were treated as equivalent, given the year-to-year variation of around 80-90 kg/ha, and the most stable of these was selected.

5. Results

5.1 Model comparison

Model	Family	Valid. MAE	Test MAE	Test MAPE (%)	Test R^2	Test R^2, excl. 2021/22 shock
Naive, same season last year	Baseline	438	561	15.5	0.38	
Naive, 3-season average	Baseline	366	532	14.8	0.43	
ARIMA(1,1,1) per district	Baseline	375	509	14.2	0.48	
Ridge regression	Weak learner	353	438	12.3	0.58	0.74
Decision tree	Weak learner	390	533	15.0	0.43	0.64
Bagging (500 trees)	Bagging	370	482	13.6	0.50	0.71
Random Forest	Bagging	368	465	13.0	0.55	0.72
XGBoost	Boosting	350	455	12.8	0.56	0.72
LightGBM	Boosting	357	463	13.0	0.55	0.71
CatBoost	Boosting	353	456	12.7	0.57	0.71
Stacking (Ridge, RF, LightGBM, CatBoost)	Stacking	350	455	12.8	0.56	0.72
Ridge + CatBoost blend	Hybrid	348	443	12.4	0.58	0.73
ARIMA + Random Forest residuals	Hybrid	369	488	13.8	0.51	0.70
Explainable Boosting Machine (selected)	Other	**347**	**446**	**12.4**	**0.58**	**0.72**
NGBoost	Other	343	459	13.0	0.54	0.72
Linear mixed-effects	Other	355	441	12.4	0.57	0.75
Gaussian Process	Other	361	426	12.0	0.60	0.74
Histogram Gradient Boosting	Other	355	458	12.9	0.56	0.71
Extra Trees	Other	361	467	12.9	0.56	0.71
Huber regression	Other	362	438	12.3	0.58	0.74
Stacked XGBoost [5]	Published	345	451	12.8	0.56	0.73
Support vector regression [6]	Published	388	443	12.2	0.59	0.70

Table 1. Validation (2010-2016) and test (2017-2026) results. MAE in kg/ha. The two published methods are a benchmark, not candidates.

- NGBoost achieved the lowest validation error (343 kg/ha), and the EBM (347 kg/ha) fell within the 1% equivalence band. The EBM was selected as the more stable of the two (standard deviation across years of 79 against 90 kg/ha).
- On the test seasons, the EBM achieved an MAE of 446 kg/ha (12.4%) and an R^2 of 0.58, a clear improvement on the naive three-season average (532 kg/ha) and ARIMA (509 kg/ha). A Wilcoxon signed-rank test found no significant difference between the EBM and ridge regression, the strongest simple model on test ($p = 0.14$), and a highly significant improvement over the naive baseline ($p < 0.001$).
- Overfitting was checked directly. The EBM's error was 235 kg/ha on training, 347 on validation and 446 on test, early stopping halted XGBoost where its validation error began to rise, and the rise on test affected every model. Half of the EBM's estimates were within 328 kg/ha (median absolute error) and its bias was -16 kg/ha. A Diebold-Mariano test on the 18 seasons confirmed the gain over the naive average ($p = 0.025$) and no significant difference from ridge regression ($p = 0.61$).
- Models with smooth or linear components (Gaussian process, ridge, Huber and mixed-effects regression) performed slightly better on the test period than tree-based models. The test period contained record yields in 2018-2020 followed by a collapse, which requires extrapolation beyond the range observed in training. The deployed model was not changed in light of these results, as selecting on the test set would invalidate it

as an independent evaluation.

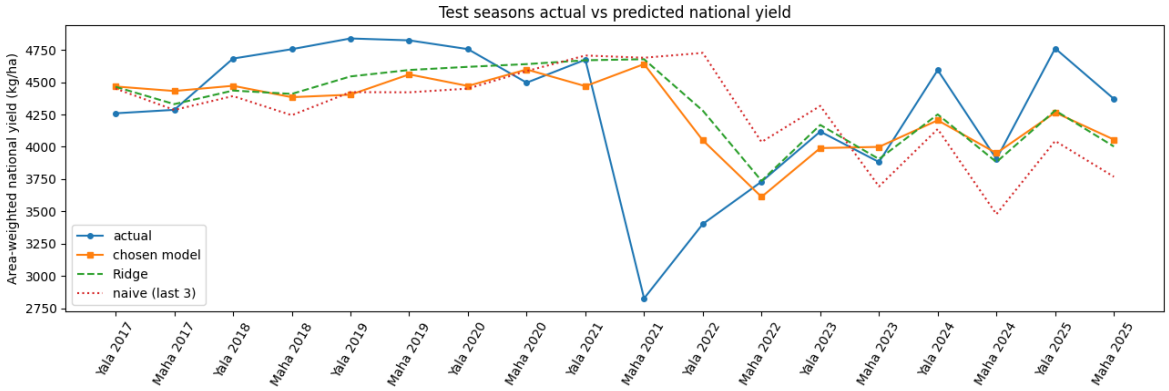


Figure 2. Area-weighted national yield in the test seasons, actual and estimated. No model anticipated the 2021/22 Maha collapse.

5.2 The 2021/22 fertilizer shock

For Maha 2021/22 the model estimated a normal season (approximately 4,640 kg/ha nationally) against an actual yield of 2,828 kg/ha, an average district error of 1,480 kg/ha. Excluding the two affected seasons, the test R^2 rises to 0.72. Errors were lowest in the large irrigated areas and in districts with stable yields (4-8%) and highest in the northern districts (14-24%), where most cultivation is rainfed and the historical record contains gaps from the conflict period.

5.3 Accuracy in planning terms

Table 2 expresses the test results in the terms a planner uses. Outside the fertilizer crisis, 93% of district estimates were within 20% of the official yield, and the implied national production was within 5.8% on average. The R^2 for production in tonnes is high mainly because larger districts produce more. It reflects the scale of the planning problem rather than a better model.

Measure (test seasons 2017-2026)	All seasons	Excluding the two ban seasons
Accuracy (100% minus mean percentage error)	87.6%	90.6%
Median percentage error	8.4%	7.4%
District estimates within 20% of official yield	86%	93%
District production (tonnes), R^2	0.95	0.98
National production per season, mean absolute error	9.5%	5.8%

Table 2. Accuracy of the selected model in planning terms. The naive three-season average placed 79% of district estimates within 20%.

5.4 Incorporating government policy

The fertilizer policy cannot be learned from 1981-2016, because no shortage occurred in that period. It was therefore tested differently. Each model was trained on all seasons up to and including Maha 2021/22, and asked to estimate Yala 2022, the second shortage season. With a yes/no shortage input, the CatBoost error fell from 814 to 318 kg/ha and its national estimate (3,474 kg/ha) came within 71 kg/ha of the actual figure. The EBM overstated the effect, as it applied the full Maha 2021/22 collapse to a milder shortage. The deployed model includes this input as a scenario switch, which a planner can use when a shortage is expected. It was not used in model selection or in any reported test score.

5.5 Uncertainty of the estimates

The intervals produced by NGBoost were found to be overconfident. Its nominal 80% interval contained the actual yield in only 46% of test seasons. A conformal range [11] was therefore adopted, derived from the 80th percentile of validation errors (±545 kg/ha). It contained the actual yield in 72% of test seasons, and in 77% when the two fertilizer-ban seasons are excluded.

5.6 Flagging a poor harvest

Planners often need a simpler signal first, so each district-season was also classified as poor (more than 10% below its three-season average), normal or good. Logistic regression, selected on the validation years by macro F1, achieved a test macro F1 of 0.50 against 0.22 for a baseline that always predicts normal, and identified 44% of poor harvests. Converting the regression estimate into a class identified fewer poor harvests (34%) but was correct 71% of the time when it did, which makes it a reliable warning. Tree-based classifiers performed poorly, because poor seasons are rare in the training years.

6. Comparison with Existing Approaches

The stacked XGBoost approach of Senevirathne and Premasiri [5] and the support vector regression approach of Ekanayake et al. [6] were reproduced on the same data and time-based split. The stacked XGBoost achieved a strong validation result (345 kg/ha) but a test MAE of 451 kg/ha, comparable to the other tree ensembles. Support vector regression was the weakest model on validation (388 kg/ha) but competitive on test (443 kg/ha), consistent with the advantage of smooth models in the test period. As the original studies used different regions, periods and inputs, their published figures are not directly comparable, so reproducing the methods under identical conditions provides the appropriate comparison.

7. Interpretation of the Model

The EBM is additive by construction. Each estimate is the sum of a baseline and one learned contribution per feature, together with a small number of pairwise terms. These contributions are its exact explanation and correspond to SHAP values [9]. SHAP, LIME [10] and the EBM's own feature functions present a consistent picture. Recent yield history is the dominant driver, followed by the irrigation mix and national conditions in the previous season. Among weather variables, the number of heavy-rain days has the clearest effect, with little impact up to about four days of 50 mm or more, followed by a progressively larger reduction. Very wet seasons, windy seasons and seasons much drier than normal also reduce the estimate. These attributions describe the model's behaviour and should not be read as causal effects.

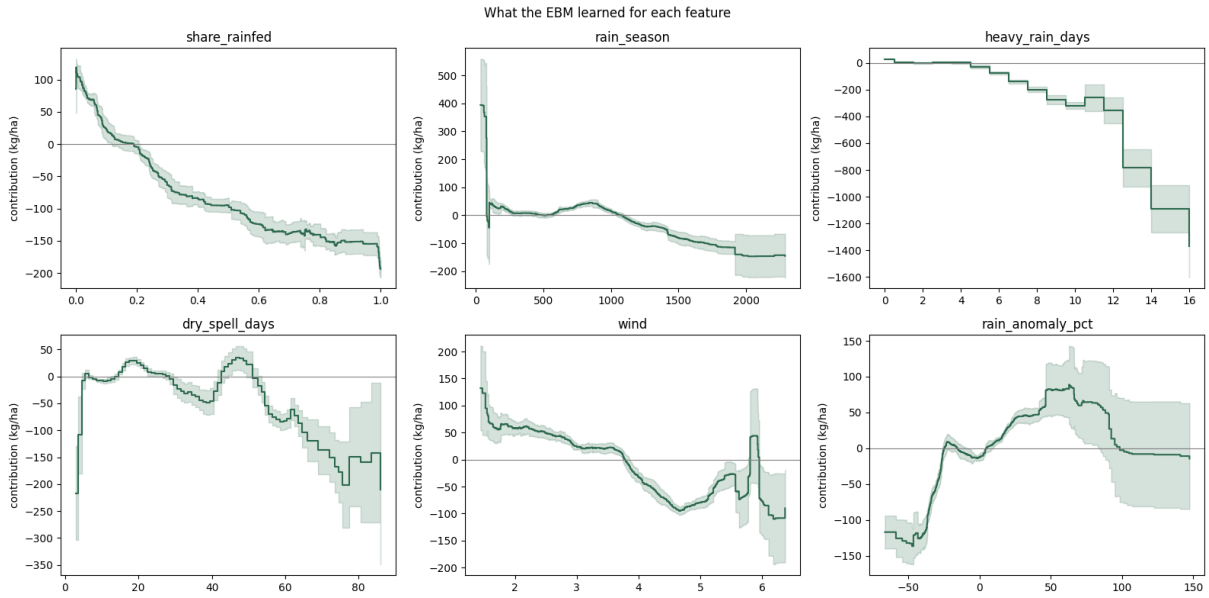


Figure 3. Contributions learned by the EBM for the rainfed share and the weather features (kg/ha), with variation across bagged fits.

SHAP dependence plots (Figure 4) quantify these effects. Each additional 100 kg/ha in a district's three-season average raises the estimate by approximately 31 kg/ha. Each additional ten percentage points of rainfed land lowers it by approximately 29 kg/ha, and seasons with eight or more heavy-rain days lower it by approximately 350 kg/ha. The explanation is also stable. When the model was retrained on random samples of 20% to 80% of the training seasons, the three-season average remained the leading feature in every case, and at least five of the

six leading features were unchanged.

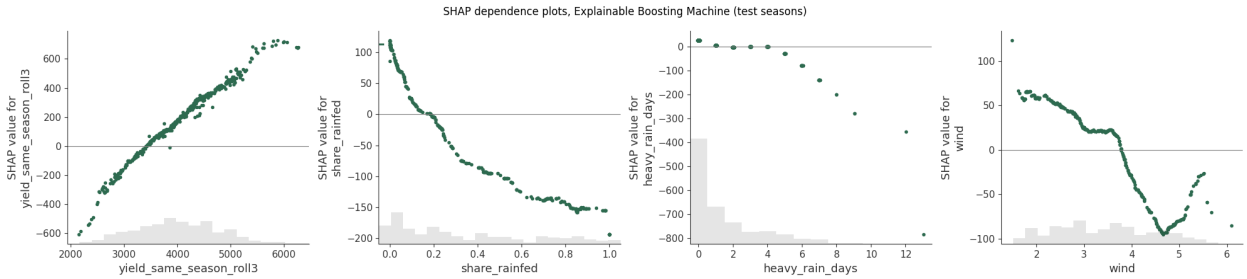


Figure 4. SHAP dependence plots for the test seasons, showing the contribution of each feature against its value.

For Polonnaruwa in Maha 2021/22 (actual 3,103 kg/ha, estimate 5,505 kg/ha), every contribution was positive, driven by the district's strong recent history. The explanation makes clear that the estimate rested on conditions that had held in every previous season and ceased to hold in 2021. For a decision maker, a large gap between the estimate and early field reports is therefore itself a warning sign.

8. Deployment

The selected model is provided through two components. An interactive Streamlit application allows a user to select a district and season, pre-fills each district's typical values, and returns the estimated yield, its likely range, a comparison with the district's history, the contribution of each input, a harvest outlook, and a fertilizer-policy scenario. Further pages vary one input across its range, map and rank district yields for any season, compare the models in three views (validation, test and the published benchmark), and estimate a whole CSV file of district-seasons. The application does not run the model itself. It sends each request to a FastAPI prediction service, which validates the input against a data contract, builds the features from the stored history, returns the estimate with its explanation through the */predict* endpoint (or many estimates through */predict/batch*), and records every prediction in a structured log. Both components are packaged as Docker containers and deployed to Azure Container Apps, and the application falls back to a local copy of the model if the service cannot be reached. On the held-out Polonnaruwa Maha 2025/26 season, the model estimated 5,026 kg/ha against an actual record yield of 6,095 kg/ha, again underestimating a sharp departure from recent history. Under the shortage scenario it would estimate 3,068 kg/ha.

The deployed web application is available at paddy-yield-app.grayflower-d27356cf.malaysiawest.azurecontainerapps.io, the prediction API at paddy-yield-api.grayflower-d27356cf.malaysiawest.azurecontainerapps.io/docs, and the complete source code, data and deployment workflows in the GitHub repository at github.com/Uthpalagit/paddy-yield-prediction-269219J. A backup copy of the application, which runs the same model file without the API, is hosted on Streamlit Community Cloud at paddy-yield-prediction-269219j-2wfd9iajbb7ebf2jpuzzgb.streamlit.app, so the application remains available if the Azure deployment is stopped.

Following standard MLOps practice, a continuous-integration workflow runs automated tests of the service and builds both containers on every change, and a scheduled monitoring job compares new inputs with the training data and, once official yields are released, the error with the validation level. Applied retrospectively to 2017-2026, the standard drift score (PSI) proved unsuitable season by season, because all districts share the same weather. The informative signal was the share of districts whose recent yields lay outside their training range, which exceeded half from Maha 2019/20 to Yala 2022, the period in which the model lagged. Error alerts fired in four seasons, and a champion-challenger rule prevented one inferior retrained model from being deployed, although retraining alone did not improve on the original model.

9. Conclusions and Recommendations

The project demonstrates that district paddy yields can be estimated at the end of each season with an accuracy of approximately 88%, several months before official figures are available, and that each estimate can be explained in terms a planner can verify. The analysis also shows that most of the predictive signal lies in yield history and irrigation, that improvements in data quality matter more than the choice among strong algorithms, and that a policy shock such as the 2021 fertilizer ban cannot be learned from historical data alone, although once observed it can be represented as a policy input.

The principal limitations are the absence of district-level data on fertilizer application, seed varieties, pests and reservoir releases, the coarse resolution of the non-rainfall weather data, and gaps in the northern districts' records. The following steps are recommended.

- Incorporate district-level fertilizer distribution data to capture input shocks and to measure their effect more precisely.
- Add reservoir storage at the start of each Yala season, if the irrigation authorities release it.
- Evaluate finer satellite imagery (such as Sentinel-2 at 10 m) over mapped paddy fields, once enough seasons exist for training.
- Evaluate a combination of the EBM with a smooth model to improve extrapolation in exceptional seasons.
- Confirm the benefit of the qualitative features, especially the El Niño phase, on the 2026/27 seasons once they are published.
- Operate the monitoring each season against the official DCS figures, treating a high share of districts beyond their training range as a signal to verify the estimates against field reports.

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